

INTERNET-OF-THINGS AND EMBEDDED SYSTEM PRACTICE

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LECTURE

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Chapter 5: IoT Architecture Models



5.1 Stages of IoT Solutions Architecture

IoT solutions involve multiple stages that collectively enable data collection, processing, and actionable insights. The stages typically include:

- **Perception Stage:**

This is the physical layer where sensors and actuators collect data from the environment. It is responsible for identifying and gathering information about physical objects such as temperature, humidity, motion, or location.

- **Network Stage:**

This stage involves transmitting the collected data to processing centers using communication networks like Wi-Fi, Bluetooth, Zigbee, cellular networks, or LPWAN. It ensures reliable and secure data transfer.

- **Processing Stage:**

Data received from the network stage is stored, processed, and analyzed here. This stage often involves cloud or edge computing platforms that transform raw data into useful information.

- **Application Stage:**

This stage provides application-specific services to users, such as smart home automation, healthcare monitoring, industrial automation, and smart cities management.

- **Business Stage:**

The final stage focuses on managing IoT systems, including business models, privacy policies, user management, and data governance to ensure sustainable and secure deployment

5.2 Three-Layer IoT Architecture

The three-layer model is a simplified IoT architecture that organizes the system into three main layers:

- **Perception Layer** (Sensing Layer):

It consists of physical sensors, actuators, RFID tags, and other embedded devices that acquire data from the environment or physical objects.

- **Network Layer:**

Responsible for transmitting data collected by the perception layer to the processing units. It includes all communication technologies and protocols such as Wi-Fi, 4G/5G, Bluetooth, Zigbee, and more.

- **Application Layer:**

This layer delivers the IoT services and applications to end-users. It provides different services tailored to specific domains like home automation, healthcare, agriculture, and transportation

Advantages:

- Simple and easy to understand
- Suitable for basic IoT implementations

Limitations:

- Lacks detailed processing and management layers
- Less scalable for complex applications

5.3 Five-Layer IoT Architecture

The five-layer architecture offers a more comprehensive IoT framework by adding two additional layers for better processing and business management:

- **Perception Layer:**

The same as in the three-layer model, responsible for sensing and data acquisition.

- **Network Layer:**

Transports data from perception devices to processing centers.

- **Processing Layer (Middleware Layer):**

Handles data storage, filtering, processing, and management. It often includes cloud computing resources, databases, and data analytics tools.



- **Application Layer:**

Provides domain-specific services to users, such as smart grids, smart healthcare, smart cities, and industrial IoT.

- **Business Layer:**

Manages overall IoT system operations including business logic, user privacy, data security, and system monitoring. It also defines the business models and policies for IoT deployment.

5.4 Cloud-Based IoT Architecture

Cloud-based IoT architecture centralizes data processing, storage, and analytics in the cloud. IoT devices send data to cloud servers where it is processed and stored.

Components:

- IoT Devices and Sensors
- Network Connectivity (Internet, Wi-Fi, Cellular)
- Cloud Servers and Storage
- Cloud Analytics and Application Services

Advantages:

- High computational power and storage capacity
- Scalability to handle large volumes of data
- Centralized management and ease of deployment
- Integration with big data analytics and AI services

Disadvantages:

- Latency issues due to distance between devices and cloud
- Dependence on stable internet connectivity
- Privacy and security concerns due to centralized data storage

5.5 Fog-Based (Edge) IoT Architecture

Fog-based, or edge computing architecture, pushes data processing closer to the data sources (i.e., at the edge of the network) rather than relying entirely on the cloud.

Components:

- IoT Devices and Sensors
- Edge/Fog Nodes (local processing units, gateways, routers)
- Network Connectivity
- Cloud Servers (for further processing if needed)

Key Features:

- Data is processed locally or near the source before sending relevant data to the cloud
- Reduces latency and bandwidth usage
- Enhances real-time processing capabilities
- Improves security by keeping sensitive data close to the source



Advantages:

- Faster response times for critical applications
- Reduced network congestion and lower bandwidth costs
- Improved privacy and security
- Supports real-time analytics and decision-making

Disadvantages:

- Requires more complex infrastructure and management
- Limited processing power compared to cloud
- Potential challenges in maintaining and updating distributed fog nodes